

Abhijeet Y. Vishwasrao

PhD Researcher · Generative AI and Foundation Models for Physics Simulation

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SUMMARY

PhD researcher building generative and foundation models for physics simulation. Works with diffusion models, flow matching and JEPA latent predictors, coupled with LLM-agent systems, to shorten engineering design cycles and speed up scientific discovery. Generated terabyte-scale datasets with open-source CFD solvers to train 100M+ parameter models on multi-GPU HPC clusters, and published in Nature Communications, NeurIPS and the Journal of Fluid Mechanics. Led collaborations across industry (RTX, CoreWeave, PhysicsX), academia (Stanford, University of Washington, Caltech) and Argonne National Laboratory.

EXPERIENCE

Graduate Researcher, University of Michigan, Ann Arbor (advisor Prof. R. Vinuesa) Aug 2025 – Present

- Building latent foundation models for parametrized PDE simulations and an agentic explorer (multi-LLM-agent system) that runs them autonomously for scientific exploration (preprint, [Agentic PDE Exploration](#), [arXiv:2604.09584](#))
- Conceived and led Policy-DRIFT (**NeurIPS 2026**, [arXiv:2605.14022](#)), first successful attempt to introduce generative AI in flow control, mentoring a master's student through implementation to publication
- Co-developed AeroJEPA [arXiv:2605.05586](#), a joint-embedding model for 3D aerodynamic fields, now in **NVIDIA PhysicsNeMo**
- Industry work with **RTX (Pratt & Whitney)** on film-cooling flow prediction, and with **CoreWeave** and **PhysicsX** on a JEPA latent foundation model for automotive aerodynamics (DrivAerML, DrivAerNet++), on a CoreWeave grant of ~41,000 A100 GPU-hours

Research Aide, Argonne National Laboratory, Lemont IL (advisors Dr. M. Min, Prof. P. Fischer) Jun – Aug 2026

- Ran NekRS conjugate heat transfer simulations over $Re = 2500\text{--}15500$ for wire-coil inserts in reactor heat exchangers, building the dataset to accelerate these simulations with latent generative models
- Built the two-stage Reynolds-decomposition decoder that recovers the turbulent fluctuations a plain autoencoder loses, with a latent predictor for unseen parameters

Doctoral Researcher (MSCA Fellow), KTH Royal Institute of Technology (advisors Profs. R. Vinuesa, H. Azizpour) Mar 2023 – Aug 2025

- Developed Diff-SPORT (**Nature Communications** 2026, [arXiv:2506.00214](#)), a diffusion-based framework for sparse reconstruction and optimal sensor placement in urban flows, validated on DNS and on experimental data of a scaled-down Beijing neighborhood
- Integrated 3D EDM-class diffusion models into **NVIDIA PhysicsNeMo** and scaled training across multi-GPU nodes
- Ran Nek5000 urban-flow simulations (5+ TB of data) and built an automated pipeline for parametrized 2D-DNS datasets

Visiting Researcher, Center for Turbulence Research, Stanford University (mentor Prof. B. J. McKeon) Jun – Jul 2024

Authored a successful proposal to the highly selective biennial CTR Summer Program and worked there on diffusion-based methods for urban flows

Master's Thesis, CEMEF, MinesParisTech, Université PSL (advisor Prof. E. Hachem) Apr – Oct 2022

Deep RL for uniform cooling in multiphase quenching simulations (**Int. J. Numer. Methods Heat & Fluid Flow** 2024, [doi:10.1108/HFF-11-2023-0713](#))

Research Trainee, CSIR-National Aerospace Laboratories, Bengaluru (mentor Dr. P. Deshpande) Jul 2019 – Jul 2020

Experiments on thrust from flexible pitching panels with PIV and unsteady force measurements (**J. Fluid Mech.** 2021, [doi:10.1017/jfm.2021.494](#))

SKILLS

ML & generative models: diffusion (EDM), flow matching, VAEs, JEPA, deep RL, surrogate modeling, LLM agents, SHAP

Programming & frameworks: Python, C++, Fortran, CUDA, PyTorch, NVIDIA PhysicsNeMo, Hydra, Docker, vLLM, AutoGen, LangChain

Solvers & HPC: Nek5000, NekRS, OpenFOAM, Cimlib, Basilisk, ANSYS Fluent; SLURM, multi-GPU training on EU and US clusters

EDUCATION

University of Michigan, PhD in Aerospace Engineering (transferred with advisor from KTH), GPA 4.0 2025 – May 2027 (expected)

KTH Royal Institute of Technology, PhD studies in Engineering Mechanics, GPA 4.0 2023 – 2025

École Polytechnique, MSc in Fluid Mechanics 2021 – 2022

Defence Institute of Advanced Technology, Pune, M.Tech in Aerospace Engineering, First Class with Distinction 2018 – 2020

Savitribai Phule Pune University, BE in Mechanical Engineering, First Class with Distinction 2014 – 2018

SELECTED PUBLICATIONS

- Vishwasrao, A.** et al. Diff-SPORT: diffusion-based sensor placement and reconstruction. *Nat. Commun.* 2026. [arXiv:2506.00214](#)
- Mahajan, A.*; **Vishwasrao, A.*** et al. Policy-DRIFT: generative flow control. *NeurIPS* 2026 (*equal contribution). [arXiv:2605.14022](#)
- Giral, F.; **Vishwasrao, A.** et al. AeroJEPA: latent representations for 3D aerodynamic fields. [arXiv:2605.05586](#), 2026
- Deshpande, P.; **Vishwasrao, A.** Kinematics and dynamics of pitching flexible panels. *J. Fluid Mech.* 2021. [doi:10.1017/jfm.2021.494](#)

HONORS & INVITED TALKS

Invited speaker, Prometheus “Accelerating the Aerospace Development Cycle: From Concept to Flight”, London Oct 2026

CoreWeave compute grant, ~41,000 A100 GPU-hours (16 GPUs, 3.5 months), on my research proposal 2026

Invited by NVIDIA to present at **GTC 2026**, San Jose (invitation-only poster session) Mar 2026

NVIDIA Certified Professional: Agentic AI (exam-based certification) Mar 2026

NVIDIA Academic Grant Program, competitive research proposal 2025

Marie Skłodowska-Curie Fellow, EU project ModelAir 2023 – 2025